

BANK DATABASE – TRIGGERS AND STORED PROCEDURES SS

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- blood_bank_db
- bookflow_db
- db1
- db2
- db3
- db4
- sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 2*

```
1 CREATE DATABASE BankDB;
2 USE BankDB;
3 CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     Email VARCHAR(100),
8     City VARCHAR(50)
9 );
10 CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
14     Balance DECIMAL(12,2) DEFAULT 0,
15     Branch VARCHAR(50),
16     Status VARCHAR(20) DEFAULT 'ACTIVE',
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	22:41:08	CREATE DATABASE BankDB	1 row(s) affected	0.016 sec
2	22:41:13	USE BankDB	0 row(s) affected	0.000 sec
3	22:41:23	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Na...	0 row(s) affected	0.047 sec
4	22:41:42	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...	0 row(s) affected	0.047 sec
5	22:42:00	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO...	0 row(s) affected	0.031 sec
6	22:42:12	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, L...	0 row(s) affected	0.031 sec

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Information

No object selected

Object Info Session

SQL File 2*

```
125 SELECT * FROM Customer;
126
127 SELECT * FROM Account;
128
129 SELECT * FROM Bank_Transaction;
130
131 SELECT * FROM Loan;
```

Result Grid

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anjali Patel	9876543215	anjali@gmail.com	Mumbai
107	Rahul Verma	9876543216	rahul@gmail.com	Delhi
108	Meena Singh	9876543217	meena@gmail.com	Pune
109	Vikram Das	9876543218	vikram@gmail.com	Kolkata

Customer 1 x Account 2 Bank_Transaction 3 Loan 4

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	22:43:52	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interes...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
11	22:44:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	22:44:00	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
13	22:44:00	SELECT * FROM Bank_Transaction LIMIT 0, 1000	20 row(s) returned	0.000 sec / 0.000 sec
14	22:44:00	SELECT * FROM Loan LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

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Navigator: SCHEMAS

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- blood_bank_db
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- sys

Administration Schemas

Information

No object selected

SQL File 2' x

Limit to 1000 rows

```
125 • SELECT * FROM Customer;
126
127 • SELECT * FROM Account;
128
129 • SELECT * FROM Bank_Transaction;
130
131 • SELECT * FROM Loan;
132
```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch	Status
10001	101	Savings	50000.00	Hyderabad	ACTIVE
10002	102	Savings	75000.00	Vijayawada	ACTIVE
10003	103	Current	120000.00	Bangalore	ACTIVE
10004	104	Savings	45000.00	Chennai	ACTIVE
10005	105	Current	90000.00	Hyderabad	ACTIVE
10006	106	Savings	60000.00	Mumbai	ACTIVE
10007	107	Savings	80000.00	Delhi	ACTIVE
10008	108	Current	110000.00	Pune	ACTIVE
10009	109	Savings	55000.00	Kolkata	ACTIVE

Customer 1 Account 2 Bank_Transaction 3 Loan 4

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	22:43:52	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interes...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
11	22:44:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	22:44:00	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
13	22:44:00	SELECT * FROM Bank_Transaction LIMIT 0, 1000	20 row(s) returned	0.000 sec / 0.000 sec
14	22:44:00	SELECT * FROM Loan LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

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No object selected

SQL File 2' x

Limit to 1000 rows

```
125 • SELECT * FROM Customer;
126
127 • SELECT * FROM Account;
128
129 • SELECT * FROM Bank_Transaction;
130
131 • SELECT * FROM Loan;
132
```

Result Grid

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-09-11 22:43:37
2	10002	DEPOSIT	15000.00	2026-09-11 22:43:37
3	10003	WITHDRAW	20000.00	2026-09-11 22:43:37
4	10004	DEPOSIT	5000.00	2026-09-11 22:43:37
5	10005	WITHDRAW	10000.00	2026-09-11 22:43:37
6	10006	DEPOSIT	8000.00	2026-09-11 22:43:37
7	10007	WITHDRAW	5000.00	2026-09-11 22:43:37
8	10008	DEPOSIT	12000.00	2026-09-11 22:43:37
9	10009	DEPOSIT	7000.00	2026-09-11 22:43:37

Customer 1 Account 2 Bank_Transaction 3 Loan 4

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	22:43:52	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interes...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
11	22:44:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	22:44:00	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
13	22:44:00	SELECT * FROM Bank_Transaction LIMIT 0, 1000	20 row(s) returned	0.000 sec / 0.000 sec
14	22:44:00	SELECT * FROM Loan LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

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No object selected

SQL File 2' x

Limit to 1000 rows

```
125 • SELECT * FROM Customer;
126
127 • SELECT * FROM Account;
128
129 • SELECT * FROM Bank_Transaction;
130
131 • SELECT * FROM Loan;
132
```

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	600000.00	7.25
506	106	Car Loan	350000.00	8.25
507	107	Personal Loan	150000.00	10.50
508	108	Home Loan	700000.00	7.00
509	109	Education Loan	180000.00	6.75

Customer 1 Account 2 Bank_Transaction 3 Loan 4 x

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	22:43:52	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
11	22:44:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	22:44:00	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
13	22:44:00	SELECT * FROM Bank_Transaction LIMIT 0, 1000	20 row(s) returned	0.000 sec / 0.000 sec
14	22:44:00	SELECT * FROM Loan LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

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MySQL Workbench

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Navigator

SCHEMAS

Filter objects

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- db1
- db2
- db3
- db4
- sys

Administration Schemas

Information

No object selected

SQL File 2' x

Limit to 1000 rows

```
492 INSERT INTO Bank_Transaction
493 (Account_No, Transaction_Type, Amount)
494 VALUES
495 (10001, 'WITHDRAW', 1000000);
496
497
```

Result Grid

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

Result 5 Result 6 Result 7 x

Read Only

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	22:50:42	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN ...	0 row(s) affected	0.000 sec
18	22:50:42	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
19	22:50:42	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN ...	0 row(s) affected	0.000 sec
20	22:50:42	CALL GetCustomerAccounts(101)	1 row(s) returned	0.016 sec / 0.000 sec
21	22:50:42	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount ...	0 row(s) affected	0.016 sec
22	22:50:42	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.000 sec
23	22:50:42	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount ...	0 row(s) affected	0.000 sec
24	22:50:42	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.000 sec
25	22:50:42	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ...	0 row(s) affected	0.015 sec
26	22:50:42	/* TEST */ UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.000 sec

Object Info Session

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Automatic context help is disabled. Use the toolbar manually get help for the current caret position or to toggle automatic help.

The screenshot shows the SQL Developer interface. On the left, the 'SCHEMAS' pane lists databases: blood_bank_db, bookflow_db, db1, db2, db3, db4, and sys. The main editor displays a SQL file with a query: `SHOW TRIGGERS;`. Below the editor is a 'Result Grid' showing a table of transactions with columns: Transaction_ID, Account_No, Transaction_Type, Amount, and Transaction_Date. The table contains 12 rows of data. To the right of the result grid is a toolbar with icons for 'Result Grid', 'Form Editor', and 'Field Types'. Below the result grid is an 'Output' window showing a list of database actions with columns: #, Time, Action, Message, and Duration / Fetch. The actions include creating procedures, calling procedures, and selecting data.

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10003	WITHDRAW	20000.00	2026-09-11 22:43:37
4	10004	DEPOSIT	5000.00	2026-09-11 22:43:37
5	10005	WITHDRAW	10000.00	2026-09-11 22:43:37
6	10006	DEPOSIT	8000.00	2026-09-11 22:43:37
7	10007	WITHDRAW	5000.00	2026-09-11 22:43:37
8	10008	DEPOSIT	12000.00	2026-09-11 22:43:37
9	10009	DEPOSIT	7000.00	2026-09-11 22:43:37
10	10010	WITHDRAW	6000.00	2026-09-11 22:43:37
11	10001	WITHDRAW	3000.00	2026-09-11 22:43:37
12	10002	DEPOSIT	5000.00	2026-09-11 22:43:37

#	Time	Action	Message	Duration / Fetch
27	22:52:29	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN Receiv...	0 row(s) affected	0.016 sec
28	22:52:29	CALL TransferMoney(10001, 10002, 5000)	0 row(s) affected	0.000 sec
29	22:52:29	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
30	22:52:29	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN ...	0 row(s) affected	0.000 sec
31	22:52:29	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec
32	22:52:29	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMA...	0 row(s) affected	0.000 sec
33	22:52:29	CALL HighBalanceAccounts(50000)	8 row(s) returned	0.000 sec / 0.000 sec

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The screenshot shows the SQL Developer interface. On the left, the 'SCHEMAS' pane lists databases: blood_bank_db, bookflow_db, db1, db2, db3, db4, and sys. The main editor displays a SQL file with a query: `SHOW TRIGGERS;`. Below the editor is a 'Result Grid' showing a table of accounts with columns: Account_No, Customer_Name, Account_Type, Balance, and Branch. The table contains 10 rows of data. To the right of the result grid is a toolbar with icons for 'Result Grid', 'Form Editor', and 'Field Types'. Below the result grid is an 'Output' window showing a list of database actions with columns: #, Time, Action, Message, and Duration / Fetch. The actions include calling procedures, selecting data, showing triggers, and creating triggers.

Account_No	Customer_Name	Account_Type	Balance	Branch
10001	Ravi Kumar	Savings	50000.00	Hyderabad
10002	Priya Sharma	Savings	75000.00	Vijayawada
10003	Arjun Reddy	Current	120000.00	Bangalore
10004	Sneha Rao	Savings	45000.00	Chennai
10005	Kiran Kumar	Current	90000.00	Hyderabad
10006	Anjali Patel	Savings	60000.00	Mumbai
10007	Rahul Verma	Savings	80000.00	Delhi
10008	Meena Singh	Current	110000.00	Pune
10009	Vikram Das	Savings	55000.00	Kolkata
10010	Divya Nair	Savings	70000.00	Kochi

#	Time	Action	Message	Duration / Fetch
35	22:52:29	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.000 sec
36	22:52:29	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
37	22:52:29	SHOW TRIGGERS	1 row(s) returned	0.000 sec / 0.000 sec
38	22:52:29	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned	0.000 sec / 0.000 sec
39	22:52:29	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	9 row(s) returned	0.000 sec / 0.000 sec
40	22:52:29	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) B...	0 row(s) affected	0.000 sec
41	22:52:29	CALL GetBranchAccounts('Hyderabad')	2 row(s) returned	0.000 sec / 0.000 sec

Navigation

SQL File 2*

Limit to 1000 rows

SHOW TRIGGERS;

Administration

Schemas

Information

No object selected

Object Info

Session

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch	Status
10001	101	Savings	50000.00	Hyderabad	ACTIVE
10002	102	Savings	75000.00	Vijayawada	ACTIVE
10003	103	Current	120000.00	Bangalore	ACTIVE
10004	104	Savings	45000.00	Chennai	ACTIVE
10005	105	Current	90000.00	Hyderabad	ACTIVE
10006	106	Savings	60000.00	Mumbai	ACTIVE
10007	107	Savings	80000.00	Delhi	ACTIVE
10008	108	Current	110000.00	Pune	ACTIVE
10009	109	Savings	55000.00	Kolkata	ACTIVE
10010	110	Savings	70000.00	Kochi	ACTIVE

Result 19Result 20Result 21Result 22Result 23Result 24Customer 25Account 26Bank_1ApplyRevertContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
45	22:52:29	CALL GetTotalCustomerBalance(101)	1 row(s) returned	0.000 sec / 0.000 sec
46	22:52:29	CREATE PROCEDURE HighBalanceAccountsQ4(IN MinimumBalance DECIMAL(10,2))	0 row(s) affected	0.015 sec
47	22:52:29	CALL HighBalanceAccountsQ4(100000)	2 row(s) returned	0.000 sec / 0.000 sec
48	22:52:29	CREATE TRIGGER PreventNegativeBalance BEFORE UPDATE ON Account F...	0 row(s) affected	0.016 sec
49	22:52:29	CREATE TRIGGER PreventInvalidTransactionAmount BEFORE INSERT ON Ba...	0 row(s) affected	0.000 sec
50	22:52:29	CREATE PROCEDURE CalculateLoanInterest(IN p_Loan_Amount DECIMAL(10,2))	0 row(s) affected	0.000 sec
51	22:52:29	CALL CalculateLoanInterest(500000, 7.5, 5)	1 row(s) returned	0.000 sec / 0.000 sec

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Navigation

SQL File 2*

Limit to 1000 rows

SHOW TRIGGERS;

Administration

Schemas

Information

No object selected

Object Info

Session

Result Grid

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-09-11 22:43:37
11	10001	WITHDRAW	3000.00	2026-09-11 22:43:37
21	10001	DEPOSIT	5000.00	2026-09-11 22:50:42
22	10001	WITHDRAW	3000.00	2026-09-11 22:50:42
23	10001	WITHDRAW	5000.00	2026-09-11 22:52:29

Result 19Result 20Result 21Result 22Result 23Result 24Customer 25Account 26Bank_1Read OnlyContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
55	22:52:29	CREATE PROCEDURE Top5Accounts() BEGIN SELECT * FROM Account...	0 row(s) affected	0.000 sec
56	22:52:29	CALL Top5Accounts()	5 row(s) returned	0.000 sec / 0.000 sec
57	22:52:29	CREATE PROCEDURE AccountCountByBranch() BEGIN SELECT Bran...	0 row(s) affected	0.000 sec
58	22:52:29	CALL AccountCountByBranch()	9 row(s) returned	0.000 sec / 0.000 sec
59	22:52:29	CREATE PROCEDURE CustomersWithHighLoans(IN p_Amount DECIMAL(10,2))	0 row(s) affected	0.000 sec
60	22:52:29	CALL CustomersWithHighLoans(300000)	8 row(s) returned	0.000 sec / 0.000 sec
61	22:52:29	CREATE TRIGGER PreventAccountDeletion BEFORE DELETE ON Account F...	0 row(s) affected	0.000 sec

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SCHEMAS

Filter objects

blood_bank_db

bookflow_db

db1

db2

db3

db4

sys

SQL File 2'

Limit to 1000 rows

SHOW TRIGGERS;

Result Grid

Filter Rows:

Transaction_IDAccount_NoTransaction_TypeAmountTransaction_Date

1710007DEPOSIT6000.002026-09-11 22:43:37

1810008WITHDRAW8000.002026-09-11 22:43:37

1910009WITHDRAW5000.002026-09-11 22:43:37

2010010DEPOSIT11000.002026-09-11 22:43:37

2110001DEPOSIT5000.002026-09-11 22:50:42

2210001WITHDRAW3000.002026-09-11 22:50:42

2310001WITHDRAW5000.002026-09-11 22:52:29

2410002DEPOSIT5000.002026-09-11 22:52:29

NULLNULLNULLNULLNULL

AdministrationSchemas

Information

No object selected

Object InfoSession

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Form Editor

Field Types

Result 19Result 20Result 21Result 22Result 23Result 24Customer 25Account 26Bank_TApplyRevertContext HelpSnippets

Output

Action Output

#TimeActionMessageDuration / Fetch

6322:52:29CALL GetTransactionHistory(10001)5 row(s) returned0.000 sec / 0.000 sec

6422:52:29CREATE PROCEDURE GetAccountSummary() BEGIN SELECT A.Acco...0 row(s) affected0.016 sec

6522:52:29CALL GetAccountSummary()10 row(s) returned0.000 sec / 0.000 sec

6622:52:29SELECT * FROM Customer LIMIT 0, 100010 row(s) returned0.000 sec / 0.000 sec

6722:52:29SELECT * FROM Account LIMIT 0, 100010 row(s) returned0.000 sec / 0.000 sec

6822:52:29SELECT * FROM Bank_Transaction LIMIT 0, 100024 row(s) returned0.000 sec / 0.000 sec

6922:52:29SELECT * FROM Transaction_Audit LIMIT 0, 1000Error Code: 1146. Table 'bankdb.transaction_audit' doesn't exist0.000 sec